TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 16

Becke3LYP results

	S-H distance in Å	H-S-H angle in degree	dipole moment in Debye	adiabatic IP in eV
LanL2DZ	1.379	94.1	1.75	10.52
6-31G	1.378	94.0	1.80	10.49
6-31G(d,p)	1.348	92.6	1.40	10.36
Experiment	1.3356	92.11	0.9766 ± 0.0003	10.449 - 10.467

The results of the calculations with LanL2DZ are very similar to that with 6-31G, i.e. the double zeta ECP basis LanL2DZ provides a really good approximation to all-electron treatment with a valence double zeta basis.

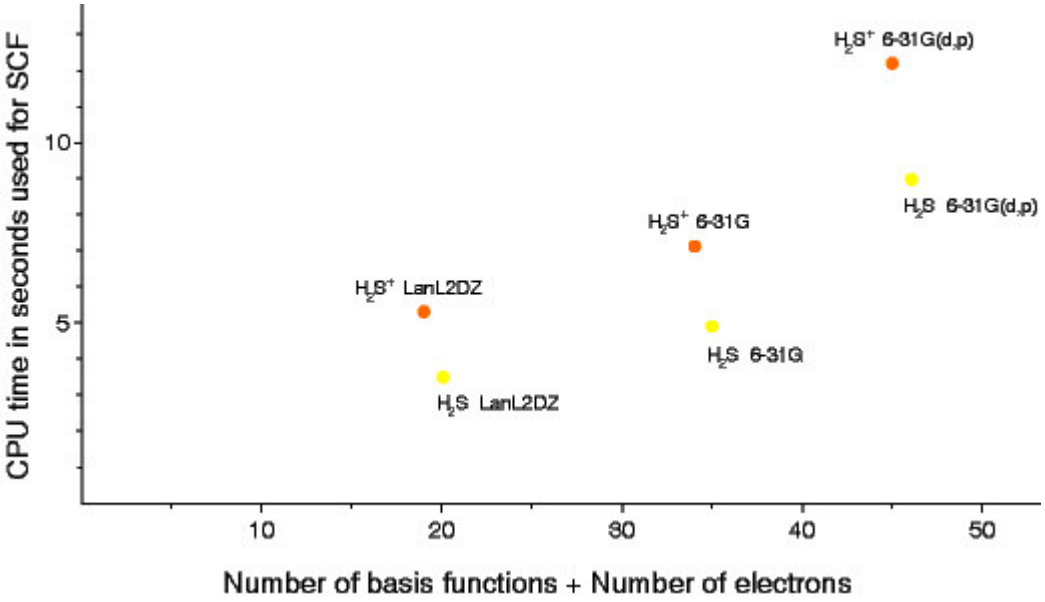
Agreement with experiment is less good since LanL2DZ does not contain polarization functions. Missing polarization functions are responsible for a large part of the differences as the 6-31G(d,p) results show.

Answers to the Questions

1. $IP_{vertical} = E_{total}(cation) - (E_{total}(neutral) + ZPE(neutral))$
2. The core levels are missing with LanL2DZ.

Supplementary Exercise: CPU Time

CPU times on a Sun Sparc Ultra Enterprise 3000



Answers to the Questions

a. The CPU time increases more than linear with model size.

b. Less electrons and less basis functions have to be taken into account.

c. H_2S^+ is open shell, H_2S closed shell. Gaussian uses by default an unrestricted SCF procedure for open shell molecules, i.e. the MOs of α and β electrons are spatially different. Therefore double number of coefficients have to be calculated and the computational expense is higher.

[General help] [Overview] [Compute] [Contact the course supervisor]

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